Next-Generation Inter-Domain Routing Architecture and Protocols

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Side Meeting@IETF 125, Shenzhen
10:00 - 12:00 Wednesday, March 18, 2026

Note Well

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Definitive information is in the documents listed below and other IETF BCPs. For advice, please talk to WG chairs or ADs:

- BCP 9 (Internet Standards Process)
- BCP 25 (Working Group processes)
- BCP 25 (Anti-Harassment Procedures)
- BCP 54 (Code of Conduct)
- BCP 78 (Copyright)
- BCP 79 (Patents, Participation)
- https://www.ietf.org/privacy-policy/(Privacy Policy)

Agenda

- **10:00-10:05 Host（5 mins）**
- **10:05-10:25 Global ROV Audit and Triangulation Tool**
  Terry Sweetser – IEISI.ORG/APNIC (20 mins)
- **10:25-10:45 The MANRS Development Process**
  Di Ma – ZDNS/MANRS Steering Committee (20 mins)
- **10:45-11:05 When RPKI Meets Reality: Understanding the causes and risks of RPKI-Invalid**
  Weitong Li – Virginia Tech (20 mins)
- **11:05-11:25 The Routing Deadlock: Challenges We Can No Longer Ignore**
  Cuicui Wang - China Unicom (20 mins)
- **11:25-11:55 QA and Discussions (30 mins)**
- **11:55-12:00 Summary/Next Steps (5 mins)**

Evolution of Inter-Domain Routing

1989

BGP1(RFC 1105)

Path Vector routing

TCP-based sessions

…

1990

BGP 2(RFC 1163)

Protocol refinements

Improved stability

…

1991

BGP 3(RFC 1267)

CIDR support

Route aggregation

…

1994

BGP 4(RFC 1771, 4271)

Current inter-domain routing standard

Multiprotocol Extension (MP-BGP, RFC 4760)

…

Ongoing

Post-BGP

Security Enhancement
(RPKI (RFC 6480), BGPsec (RFC 8205))
Observability
(BMP (RFC 7854))
Core Protocol Extensions
(FlowSpec, BGP-LS, Communities…)

Limitations of Inter-Domain Routing (BGP)

Security

- The risks in inter-domain routing primarily stem from inherent design limitations in inter-domain routing protocols. In particular, security was not treated as a first-class consideration in the design of BGP.
- The absence of a built-in trust model has resulted in persistent routing security issues and has further impeded effective coordination and optimization of routing policies.

Manageability

- Misconfigurations in BGP by network operators frequently lead to service disruptions.
- Inter-domain routing systems often exhibit delayed anomaly detection and high recovery latency, resulting in prolonged periods of network instability.

Scalability

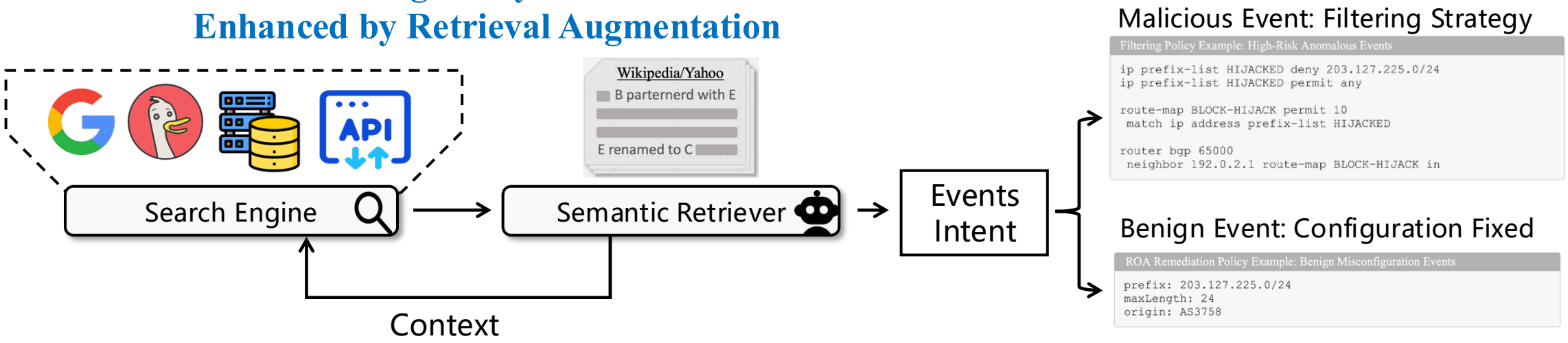
- With the continuous expansion of the Internet, the rapid growth of routing table size is imposing increasing pressure on the performance of inter-domain routing systems.
- The rigid and static semantics of BGP limit its ability to adapt to the evolving demands of emerging network services.

Exploring Agentic Inter-Domain Routing

- Addressed Problem: Routing anomalies exhibit diverse and complex characteristics.

- An iterative framework that leverages multi-round knowledge retrieval and trusted evidence fusion to progressively capture the contextual semantics of routing anomalies and infer interpretable event intents.
- Based on the inferred intents, differentiated routing policies are adaptively generated, ensuring semantic consistency between anomaly understanding and policy decisions.

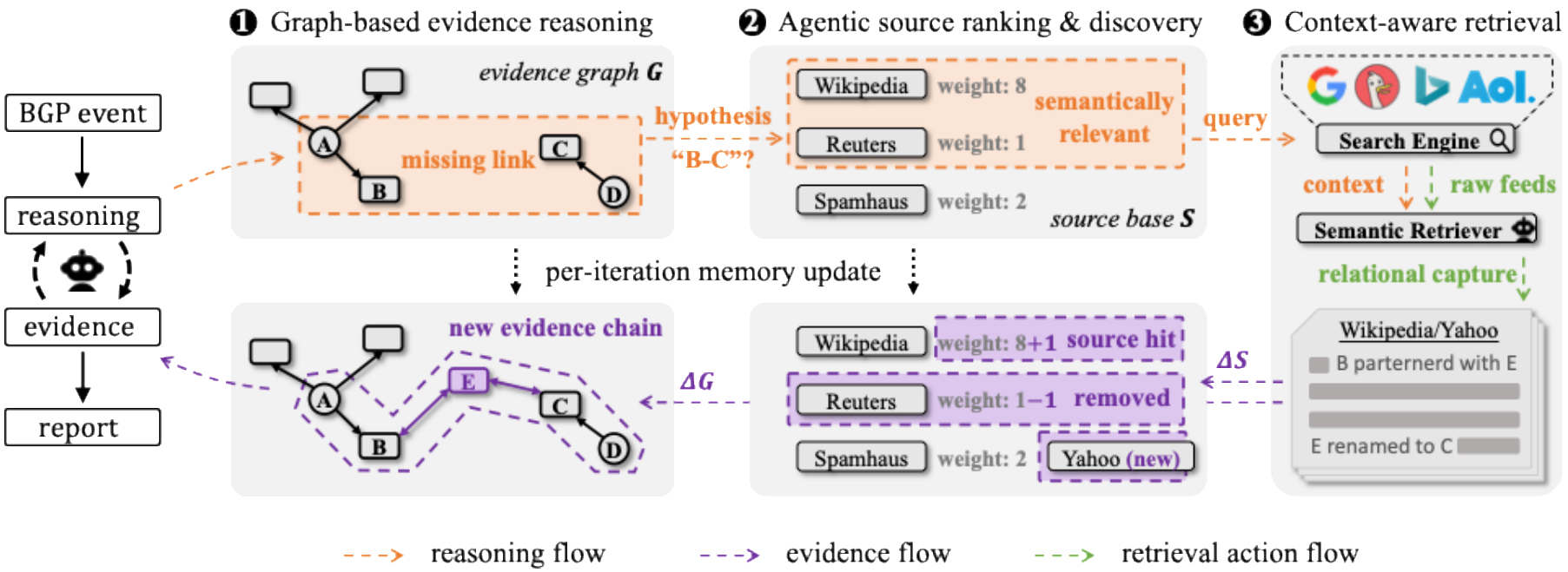
Intent-Aware Routing Policy Generation for Events Enhanced by Retrieval Augmentation



Agentic Inter-Domain Routing Policy Engine

Core Design: ReAct Agent Driven Knowledge Discovery and Decision-Making

- Multi-Source Knowledge Management: Efficient acquisition and reliability-aware evaluation of routing knowledge
- Knowledge Graph-Enabled Event Reasoning: Interpretable discovery of evidence chains for routing incidents
- Intent-Aware Agentic Routing Policy: Adaptive mitigation strategies for diverse routing anomalies



Preliminary Results

Event Intent Recognition

Event Type	Number	Percentage	Routes	Source ASes	Vantage Points	Countries Involved
High-Risk Anomalous	12	26.7%	102	22	34	12
Benign Misconfiguration	33	73.3%	331	29	50	15
Total	**45**	**100%**	**433**	**51**	**50**	**24**

Validation Set

Model	TP	TN	Accuracy	Precision	Recall	F1
Claude Sonnet 4.6	10	32	0.933	0.909	0.833	0.870
GPT-5.2	10	30	0.889	0.833	0.833	0.833
Gemini 3.1 Pro Preview	8	31	0.867	0.800	0.667	0.727
Qwen3 Max Thinking	10	30	0.889	0.833	0.833	0.833
DeepSeek V3.2	8	30	0.844	0.800	0.667	0.727

Event Intention Recognition Results

Various models have achieved good performance on all indicators, with the highest F1 value reaching 0.870, corresponding to approximately 93.3% discrimination accuracy

Response Policy Actionability

Across all correctly identified events, generated policies achieve full syntactic and semantic validity

Event Type	Policy
High-Risk Anomalous	Route Filtering
Benign Misconfiguration	ROA Remediation

Metric	Successes	Success Rate
Syntactic Correctness	42	100%
Semantic Consistency	42	100%

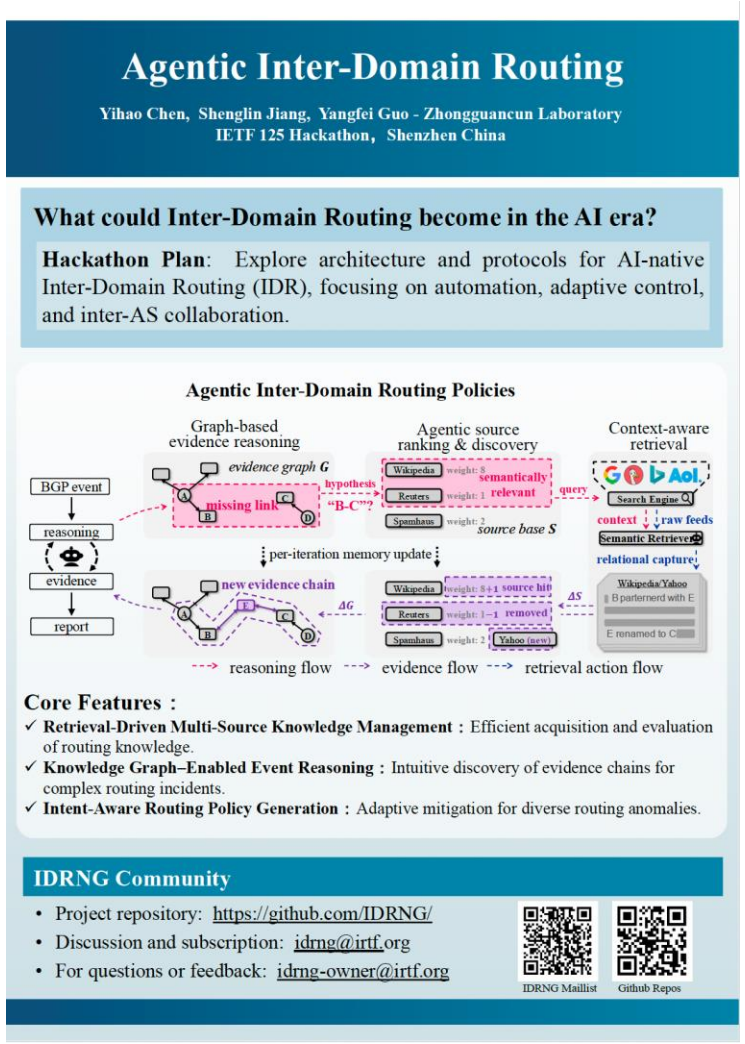
Filtering Policy Example: High-Risk Anomalous Events

```
ip prefix-list HIJACKED deny 203.127.225.0/24
ip prefix-list HIJACKED permit any

route-map BLOCK-HIJACK permit 10
 match ip address prefix-list HIJACKED

router bgp 65000
 neighbor 192.0.2.1 route-map BLOCK-HIJACK in
```

ROA Remediation Policy Example: Benign Misconfiguration Events

```
prefix: 203.127.225.0/24
maxLength: 24
origin: AS3758
```

IETF 125 Hackathon



Meeting Materials

Next-Generation Inter-Domain Routing Architecture and Protocols (IDRNG)

- **Github repos**
  https://github.com/IDRNG
- **Mailing list**
  idrng@irtf.org
- **IDRNG-IETF125**
  https://github.com/IDRNG/125meeting